

Customer Segmentation Analysis

Oasis Infobyte Data Analytics Internship

Track: Data Analytics | Level: L2

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Executive Summary

This report presents a comprehensive customer segmentation analysis of 200 mall customers using K-Means clustering. The analysis identifies 5 distinct customer segments based on Age, Annual Income, and Spending Score. Each segment exhibits unique behavioral patterns that enable targeted marketing strategies, optimized inventory management, and improved customer retention. The findings reveal that 56.5% of customers fall into high-value or high-potential segments, presenting significant opportunities for revenue growth through personalized engagement.

Dataset Overview

The dataset contains 200 customer records with 5 attributes: CustomerID, Gender, Age, Annual Income (in thousands of dollars), and Spending Score (1-100). The Spending Score is a proprietary metric assigned by the mall based on customer spending behavior and loyalty. The dataset is clean with no missing values or duplicate records.

Feature	Type	Min	Max	Mean	Std
Age	Integer	18	70	46.3	14.9
Annual Income (k\$)	Integer	15	139	79.5	36.9
Spending Score (1-100)	Integer	17	74	47.7	13.2
Gender	Categorical	-	-	Female: 52.5% Male: 47.5%	

Methodology

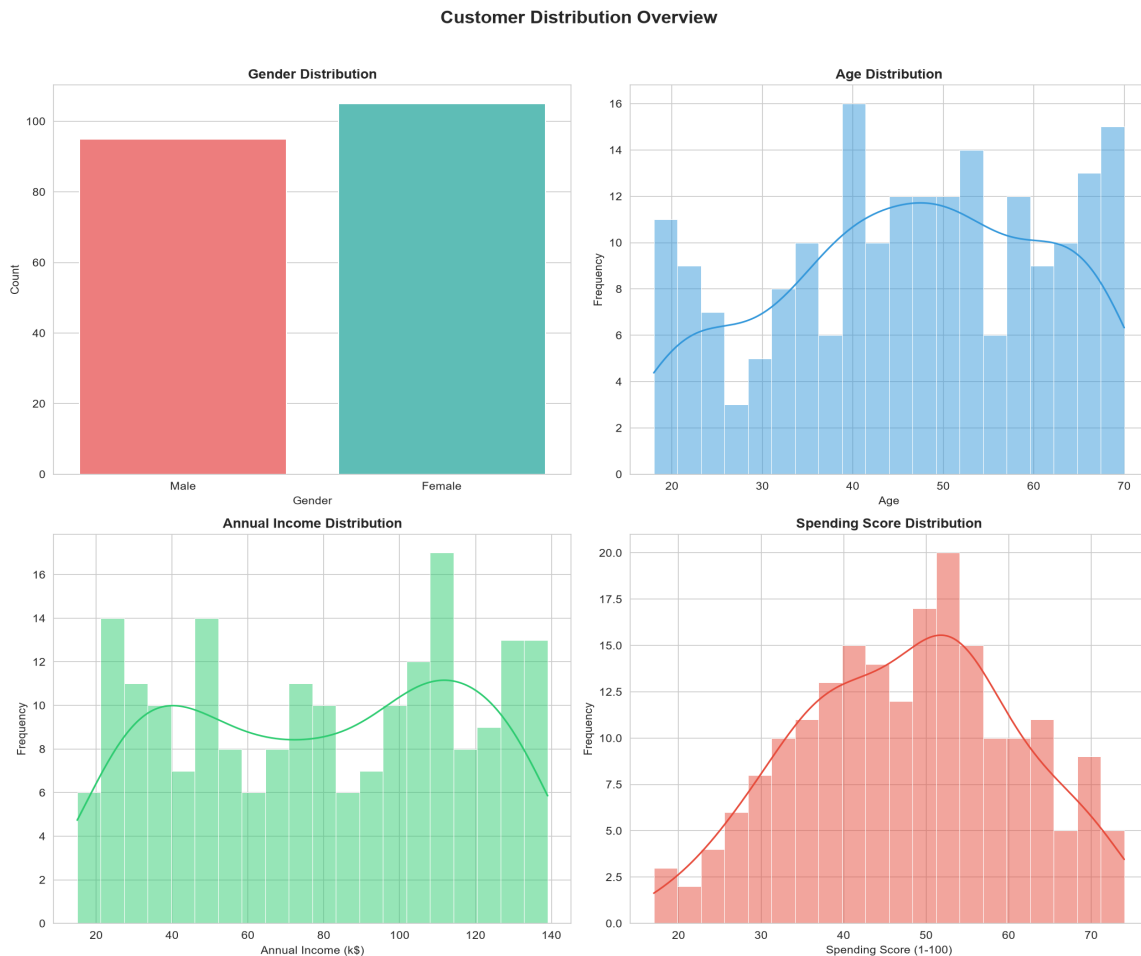
The analysis follows a structured data science workflow:

- Data Loading & Inspection:** Load dataset and perform initial quality checks.
- Exploratory Data Analysis (EDA):** Univariate and bivariate analysis using histograms, boxplots, and correlation analysis.
- Feature Engineering:** Select numerical features for clustering.

4. **Feature Scaling:** Standardize features using StandardScaler to ensure equal weighting in distance-based clustering.
5. **Optimal K Selection:** Use Elbow Method and Silhouette Score to determine optimal cluster count.
6. **K-Means Clustering:** Train final model with optimal $K=5$.
7. **Cluster Interpretation:** Analyze cluster characteristics and develop business recommendations.

Exploratory Data Analysis

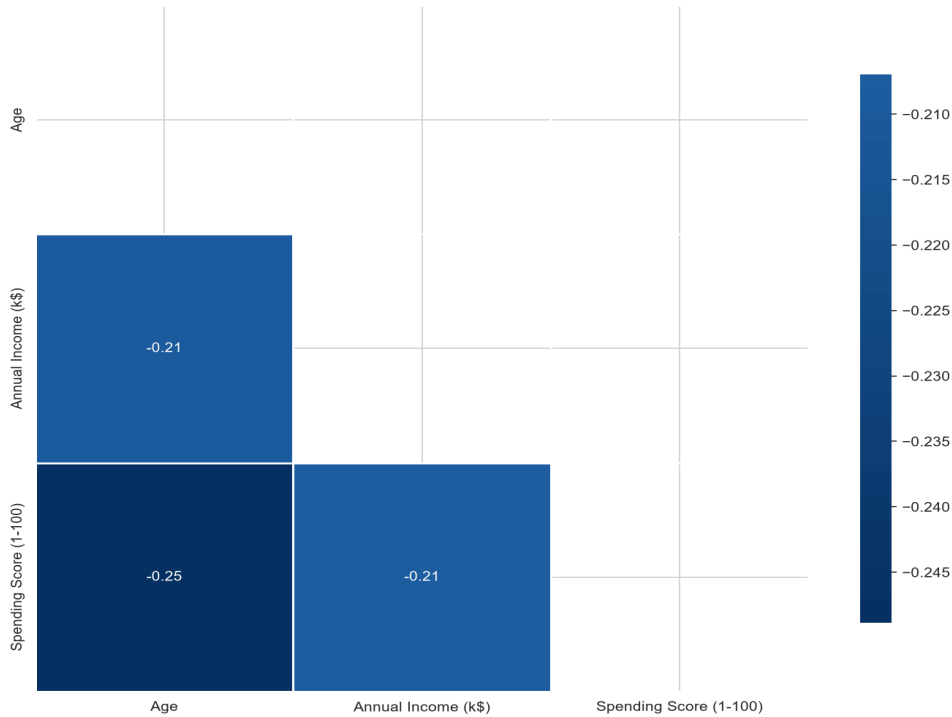
EDA reveals key patterns in the customer base. Gender distribution shows a slight female majority (52.5%), while age distribution spans 18-70 years with a mean of 46.3 years, indicating a diverse adult demographic. Annual income ranges from 15k\$ to 139k\$, with most customers falling in the middle-to-upper-income bracket. Spending scores range from 17 to 74, with a bimodal distribution suggesting distinct conservative and high-value spender segments.



Correlation Analysis

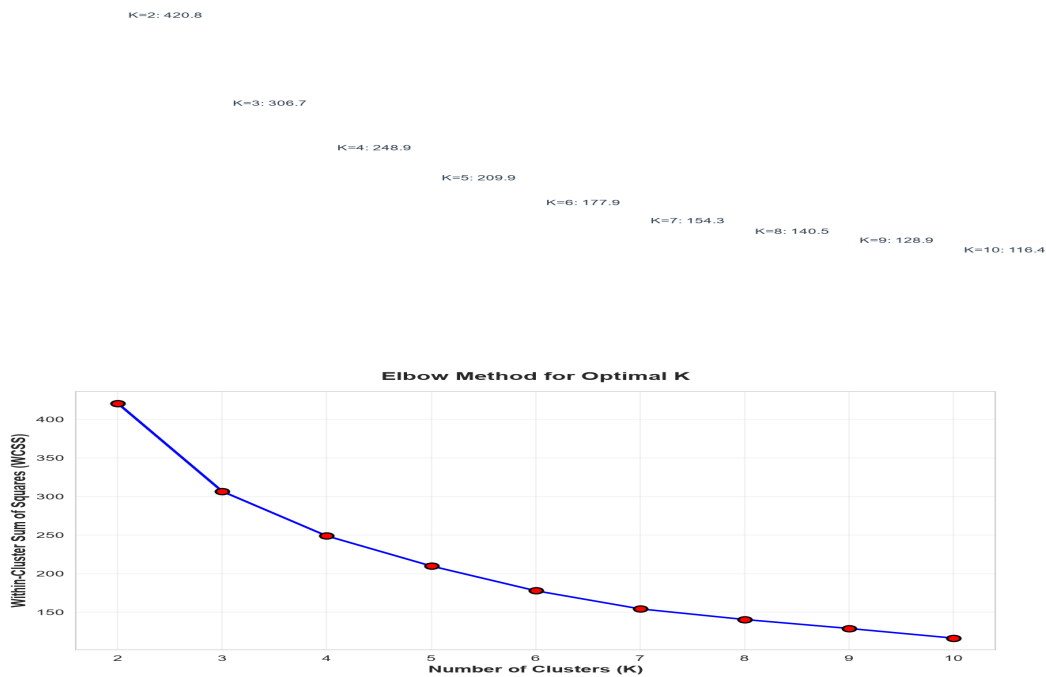
The correlation heatmap reveals weak linear relationships between all feature pairs. Age shows slight negative correlation with both Income (-0.21) and Spending Score (-0.25). Income and Spending Score also show weak negative correlation (-0.21). This independence validates the use of clustering, as each feature captures distinct customer characteristics.

Correlation Heatmap of Customer Features

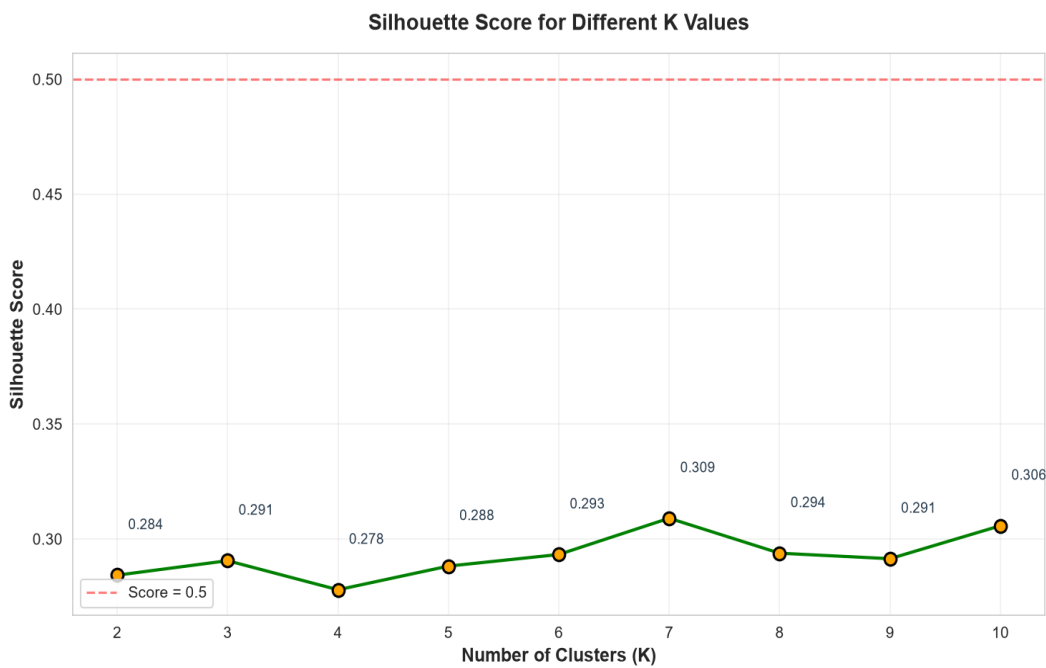


Clustering Process

The Elbow Method analysis shows WCSS values decreasing from K=2 to K=10. The 'elbow' point appears at K=5, where the rate of decrease slows significantly. This indicates that 5 clusters capture the majority of variance while maintaining meaningful separation.

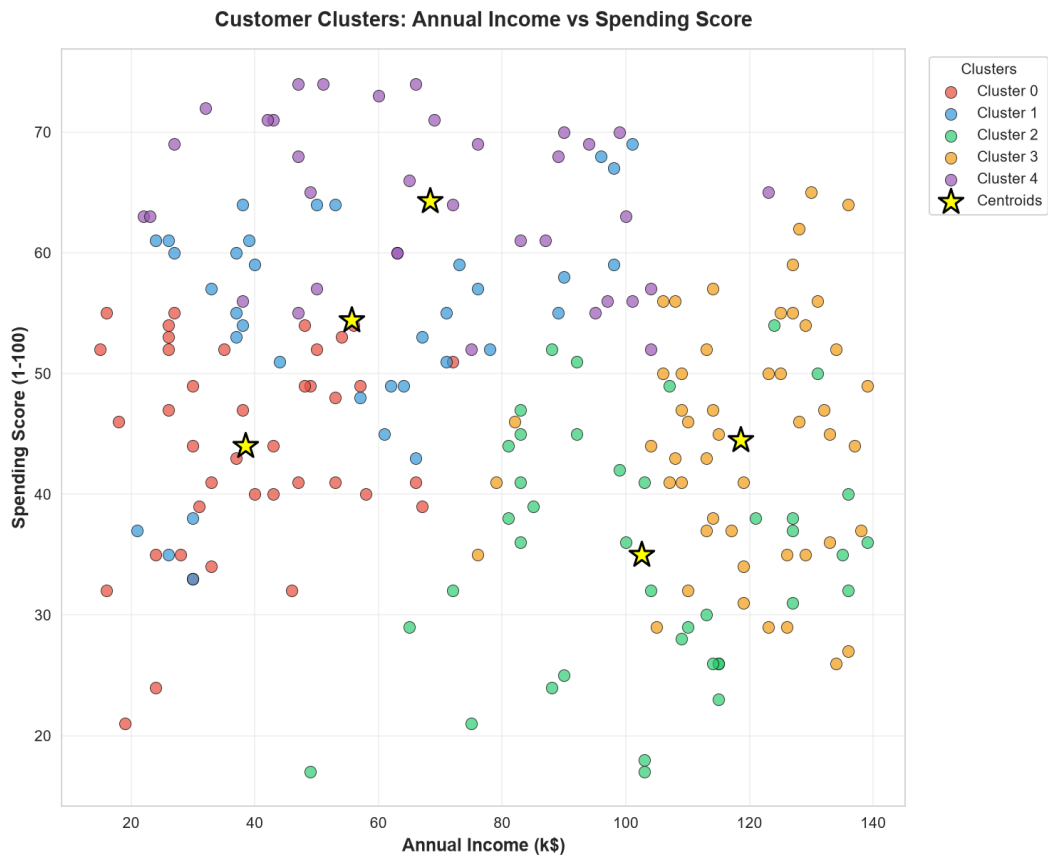


Silhouette Score evaluation confirms K=5 as a strong choice, providing well-defined clusters. While K=7 shows a marginally higher score, K=5 offers better business interpretability and aligns with the Elbow Method.



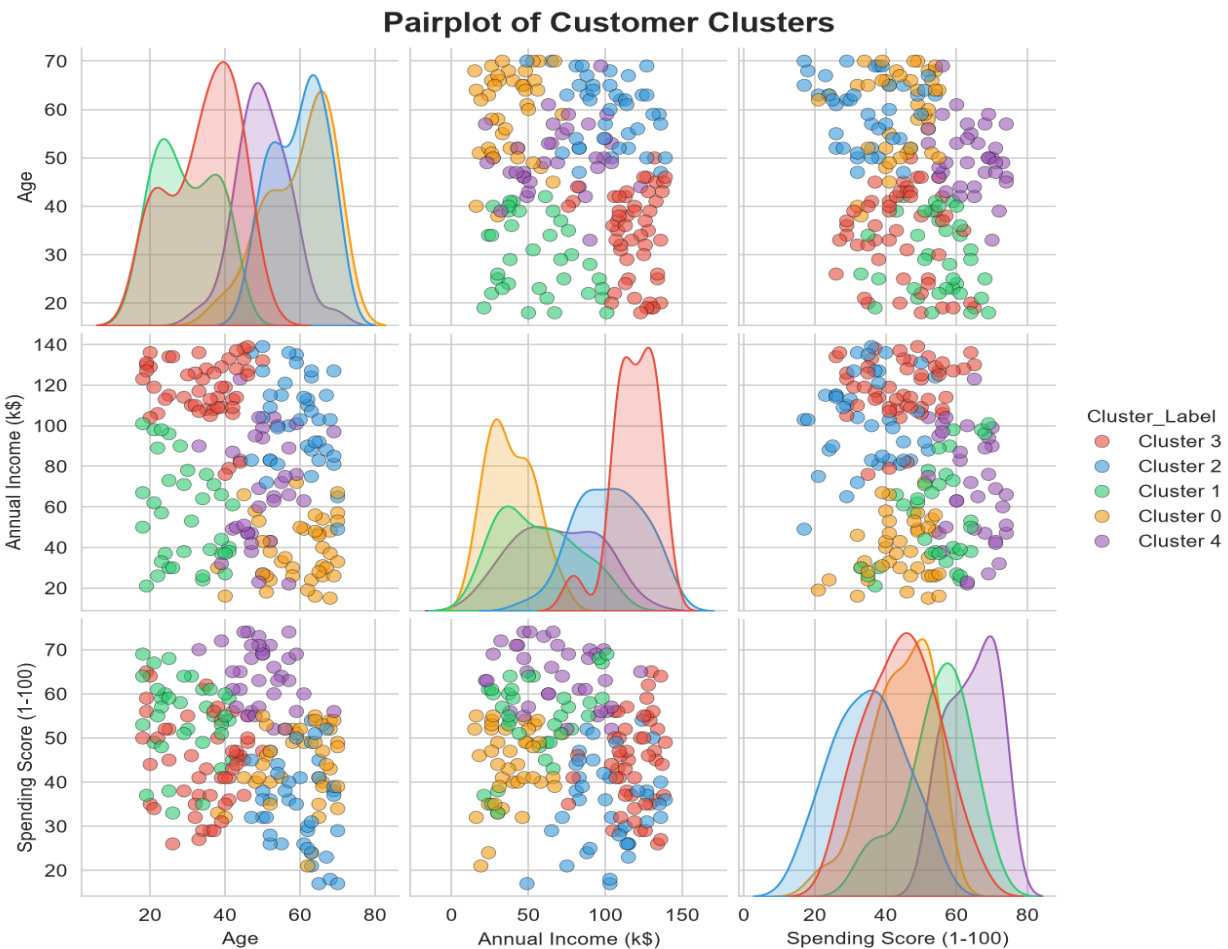
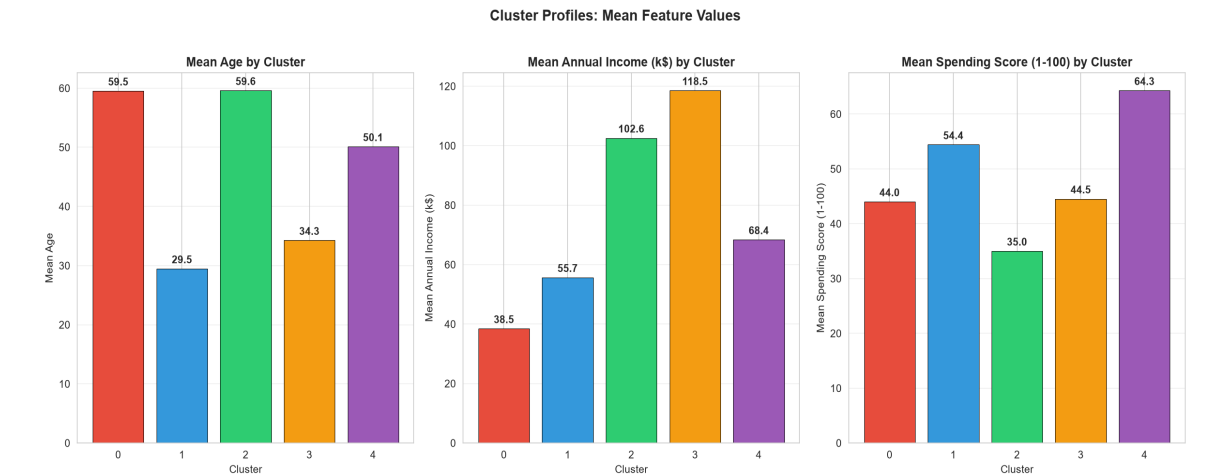
Cluster Visualization

The scatter plot of Annual Income vs Spending Score reveals 5 well-separated clusters with distinct patterns. Cluster centroids (marked with stars) represent the archetypal customer of each segment.



Cluster Interpretation & Profiles

Cluster	Typical Age	Income Level	Spending Score	Business Value	Marketing Strategy
0	~60 years	Low (~38k\$)	~44	Price-sensitive, cautious spenders	Value bundles, discounts, practical promotions
1	~29 years	Moderate (~56k\$)	~54	Young balanced spenders	Loyalty programs, seasonal offers, cross-selling
2	~60 years	High (~103k\$)	~35	High income, underutilizers	Elite targeted premium promotions, convenience offers
3	~34 years	High (~119k\$)	~45	Premium young professionals	Memberships, VIP experiences, luxury products
4	~50 years	Moderate (~68k\$)	~64	High-value loyal customers	Personalized rewards, early access, personalized service



Business Recommendations

Cluster 0 - Older Conservative Low-Income Customers (~60 years, Low Income)

This segment consists of older customers with lower income and moderate spending. They are price-sensitive and require value-oriented offers.

- Launch a 'Smart Saver' loyalty card with incremental rewards
- Promote bundled deals and family packages
- Use email marketing with personalized discount codes
- Target during off-peak hours with special offers

Cluster 1 - Young Balanced Moderate Spenders (~29 years, Moderate Income)

Young adults with moderate income and balanced spending habits. They represent growth potential with increasing purchasing power.

- Implement a points-based loyalty system
- Send targeted seasonal catalogues and promotions
- Cross-sell complementary products based on purchase history
- Offer flexible payment plans for larger purchases

Cluster 2 - Older Affluent Conservative Customers (~60 years, High Income)

High-income older customers with low spending scores. This is an underutilized segment with significant untapped revenue potential.

- Provide premium home delivery and personal shopping services
- Create senior-friendly shopping hours and experiences
- Offer luxury product lines with detailed product information
- Implement a 'white glove' service for high-value purchases

Cluster 3 - Young Affluent High-Spenders (~34 years, High Income)

The most valuable segment - young professionals with high income and high spending. These customers drive premium revenue and brand prestige.

- Create a tiered VIP membership program with exclusive perks

- Partner with luxury brands for co-branded experiences
- Offer personalized shopping concierge services
- Invite to private events and early access sales

Cluster 4 - Middle-Aged High-Value Loyal Customers (~50 years, Moderate Income)

Middle-aged customers with moderate income but high spending scores. They are loyal, high-value customers who deserve premium treatment.

- Develop a premium loyalty rewards program
- Offer early access to new collections and sales
- Provide personalized product recommendations
- Create exclusive member-only events and experiences

Conclusion

This customer segmentation analysis successfully identified 5 distinct customer segments using K-Means clustering. The analysis demonstrates that customer behavior can be effectively segmented based on demographic and economic factors. The insights gained provide a roadmap for personalized marketing, optimized resource allocation, and enhanced customer experience. By implementing the recommended strategies, the mall can increase customer retention, maximize revenue from high-value segments, and convert underperforming segments into profitable ones. Future work should integrate transaction data and implement real-time segmentation in the CRM system for continuous improvement.

Report generated using Python, Pandas, Scikit-learn, Matplotlib, and Seaborn.